



Housing prices as a benchmark for us economy

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Creating an Econometric Model

STEP 1

Equation

$$\text{Land Price} = \beta_1 + \beta_2(\text{GDP Growth}) + \beta_3(\text{Interest Rate}) + \beta_4(\text{Income}) + \beta_5(\text{Inflation}) + \beta_6(\text{Unemployment}) + \varepsilon$$

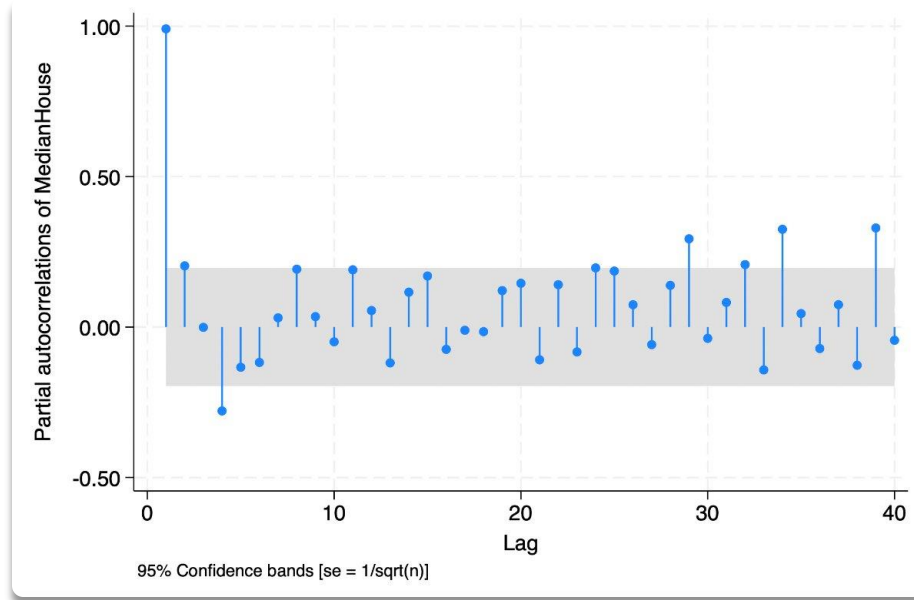
There is **one endogenous variable**, **Land Price**, and **five exogenous variables**: **GDP**, **Interest Rate**, **Real Income**, **Inflation**, and **Unemployment**. The model also includes **one error term** (ε).

An **endogenous variable** is a variable whose value is **determined by the model itself**.

In other words, it's a variable that is **explained by other variables** in the system.

An **exogenous variable** is a variable whose value is **determined outside the model** and is **not explained** by other variables in the system. It is assumed to **influence the outcome**, but **not be influenced by it**.

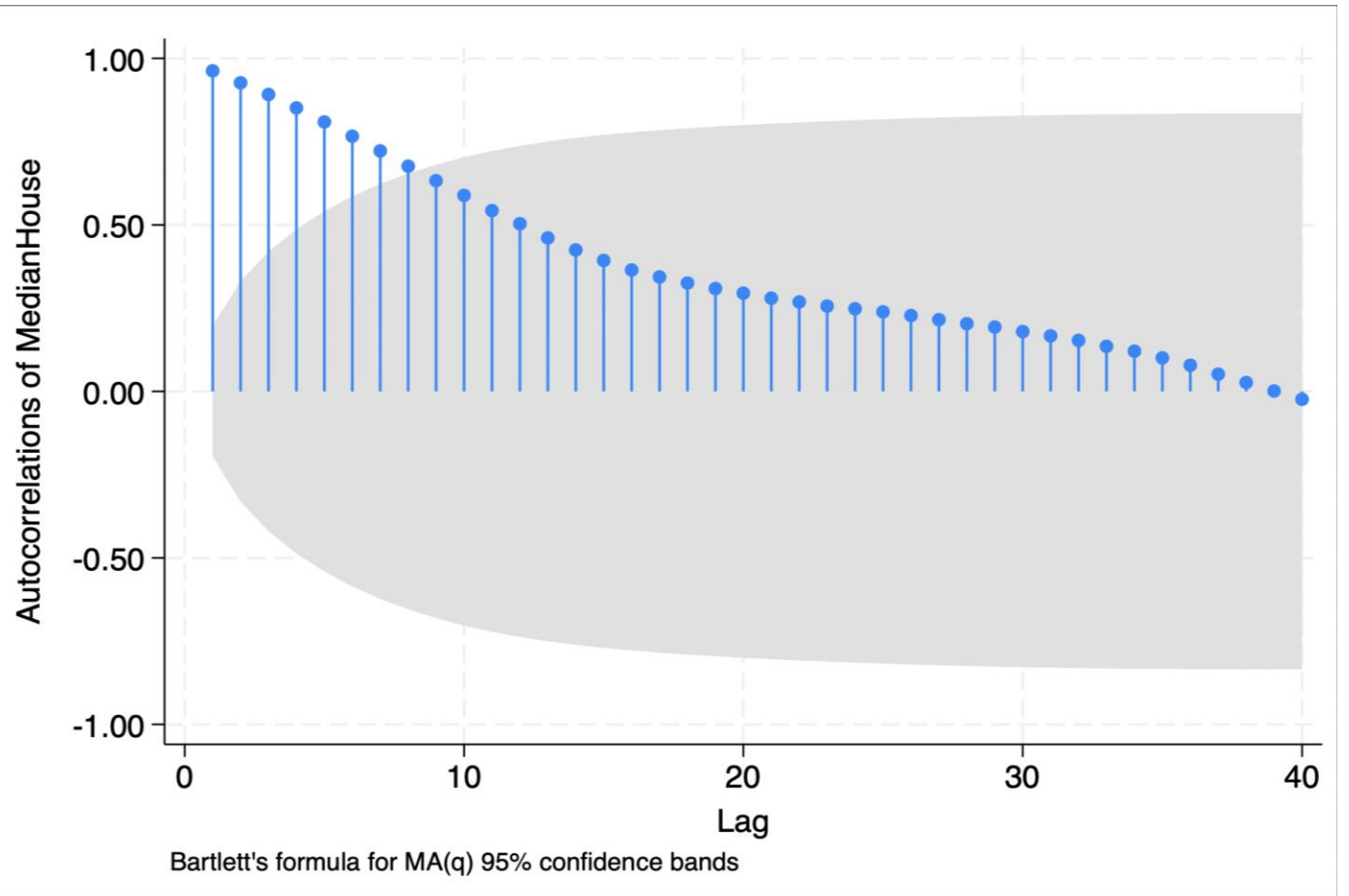
Partial Autocorrelation



- ▶ Lag 1 has a very strong and significant spike
- ▶ This suggests a strong relationship between current house prices and prices from the previous period.
- ▶ Lags 2 and beyond fluctuate within the 95% confidence band, with no clearly significant spikes.
 - ▶ This tells us we can use an Auto recursive model with one or possibly two lags

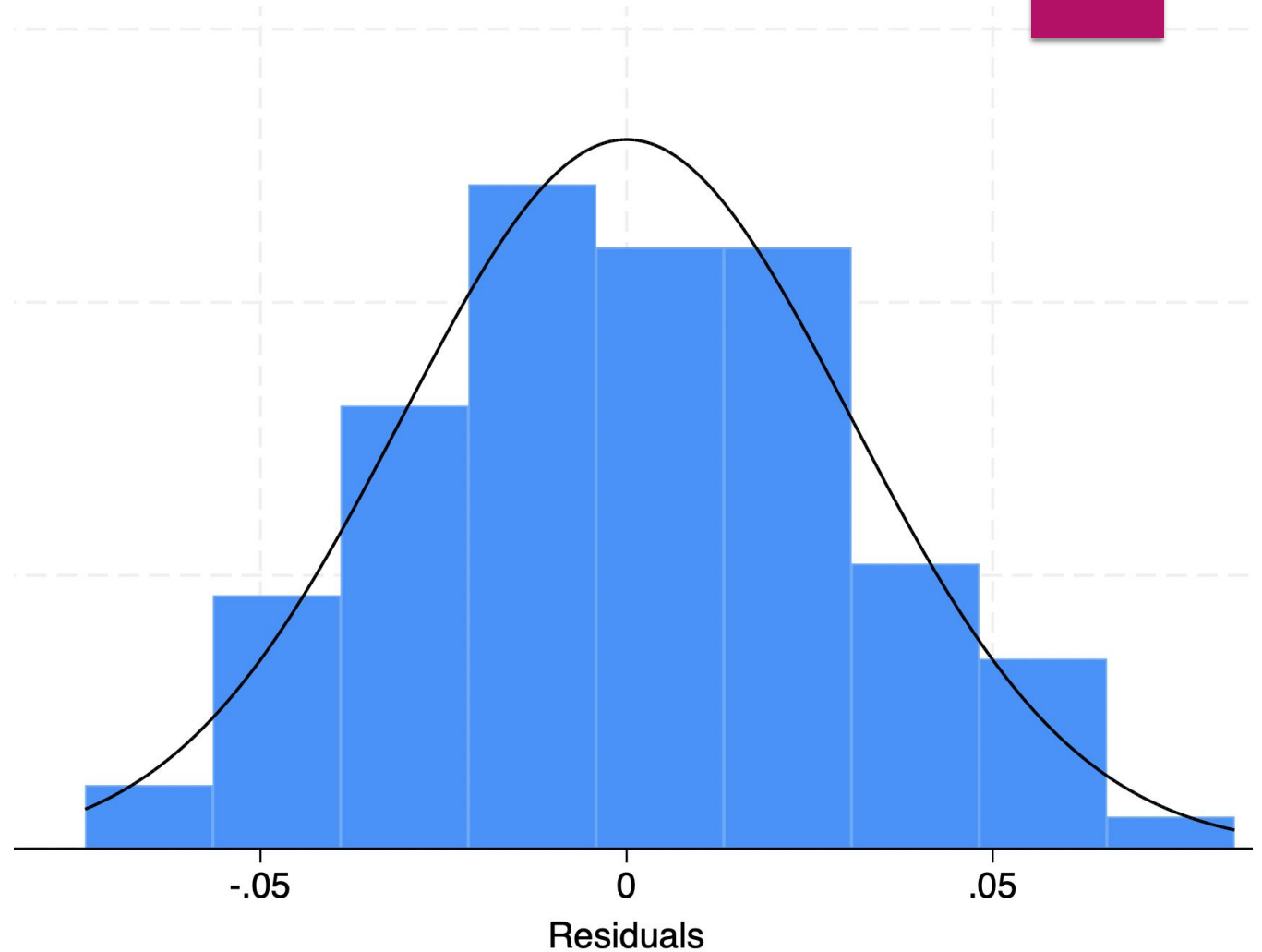
Correlation Function

- ▶ Auto Regressive 1 Model
- ▶ $y_t = \alpha + \beta_1 y_{t-1} + \epsilon_t$
- ▶ Auto regressive 2 model
- ▶ $y_t = \alpha + \beta_1 y_{t-1} + \beta_2 y_{t-2} + \epsilon_t$



Residuals Test

- ▶ Residual Distribution (Bell Curve)
- ▶ Residuals appear approximately normally distributed
- ▶ Supporting the model's assumptions of homoscedasticity and unbiasedness.



Testing Goodness of Fit

- ▶ The regression on the right explains:
- ▶ Goodness of fit
- ▶ How statistically significant each variable is
- ▶ How well the model fits with historical data, which will be used for forecasting
- ▶ We observe an R-squared of 0.95, meaning the model fits very well with past data.
- ▶ Out of the four independent variables:
- ▶ Real GDP and Real Interest Rate are statistically significant
- ▶ Unemployment Rate is partially significant
- ▶ Disposable Income is not statistically significant

```
. reg MedianHousePrice RealGDP YRIR RealDispInc UnempRate
```

Source	SS	df	MS	Number of obs	=	96
				F(4, 91)	=	540.92
Model	5.2012e+11	4	1.3003e+11	Prob > F	=	0.0000
Residual	2.1875e+10	91	240387721	R-squared	=	0.9596
				Adj R-squared	=	0.9579
Total	5.4199e+11	95	5.7052e+09	Root MSE	=	15504

MedianHous~e	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
RealGDP	27.36538	3.464804	7.90	0.000	20.48298	34.24779
YRIR	8195.817	2858.681	2.87	0.005	2517.397	13874.24
RealDispInc	1.66997	3.894857	0.43	0.669	-6.066685	9.406625
UnempRate	-2094.298	1244.944	-1.68	0.096	-4567.227	378.631
_cons	-236821.7	24851.31	-9.53	0.000	-286185.8	-187457.6

Testing Goodness of Fit

- ▶ The regression on the right is a Dickey-Fuller Test, used to determine whether the data is stationary or non-stationary.
- ▶ The issue with the first regression is that it may produce a misleading R-squared if the data is non-stationary.
- ▶ That model assumes stationarity — meaning the data does not trend up or down over time.
- ▶ The Dickey-Fuller test reports an R-squared of 0.118, which is much lower than the 0.95 in the first regression.
- ▶ This lower R-squared is acceptable because the model is now correctly accounting for the lack of stationarity in the data.

Source	SS	df	MS	Number of obs	=	95
				F(4, 90)	=	3.01
Model	943689313	4	235922328	Prob > F	=	0.0221
Residual	7.0503e+09	90	78336333.9	R-squared	=	0.1181
				Adj R-squared	=	0.0789
Total	7.9940e+09	94	85042120.9	Root MSE	=	8850.8

x1	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
x2	1.824283	8.208943	0.22	0.825	-14.48421	18.13278
x3	6183.485	3713.644	1.67	0.099	-1194.317	13561.29
x4	1.01554	3.121386	0.33	0.746	-5.185638	7.216717
x5	-2068.787	1940.485	-1.07	0.289	-5923.899	1786.325
_cons	2397.111	1240.03	1.93	0.056	-66.42584	4860.648

Log Regression

- ▶ We see a higher r-squared due to the nature of the logged values
- ▶ A one percent increase change.
- ▶ Increase decrease interest rate by -6 percent

Source	SS	df	MS	Number of obs	=	100
Model	7.1444216	5	1.42888432	F(5, 94)	=	676.34
Residual	.198591266	94	.002112673	Prob > F	=	0.0000
Total	7.34301286	99	.074171847	R-squared	=	0.9730
				Adj R-squared	=	0.9715
				Root MSE	=	.04596

MedianHouse	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
IR	-.0062989	.0055655	-1.13	0.261	-.0173494	.0047516
Inflation	.1966653	.0674938	2.91	0.004	.0626548	.3306759
UNEMP	-.0178097	.003252	-5.48	0.000	-.0242667	-.0113528
GDPln	.7153314	.0363035	19.70	0.000	.64325	.7874129
Dispincln	.0709509	.0164111	4.32	0.000	.0383664	.1035355
_cons	4.881442	.2666577	18.31	0.000	4.351987	5.410897

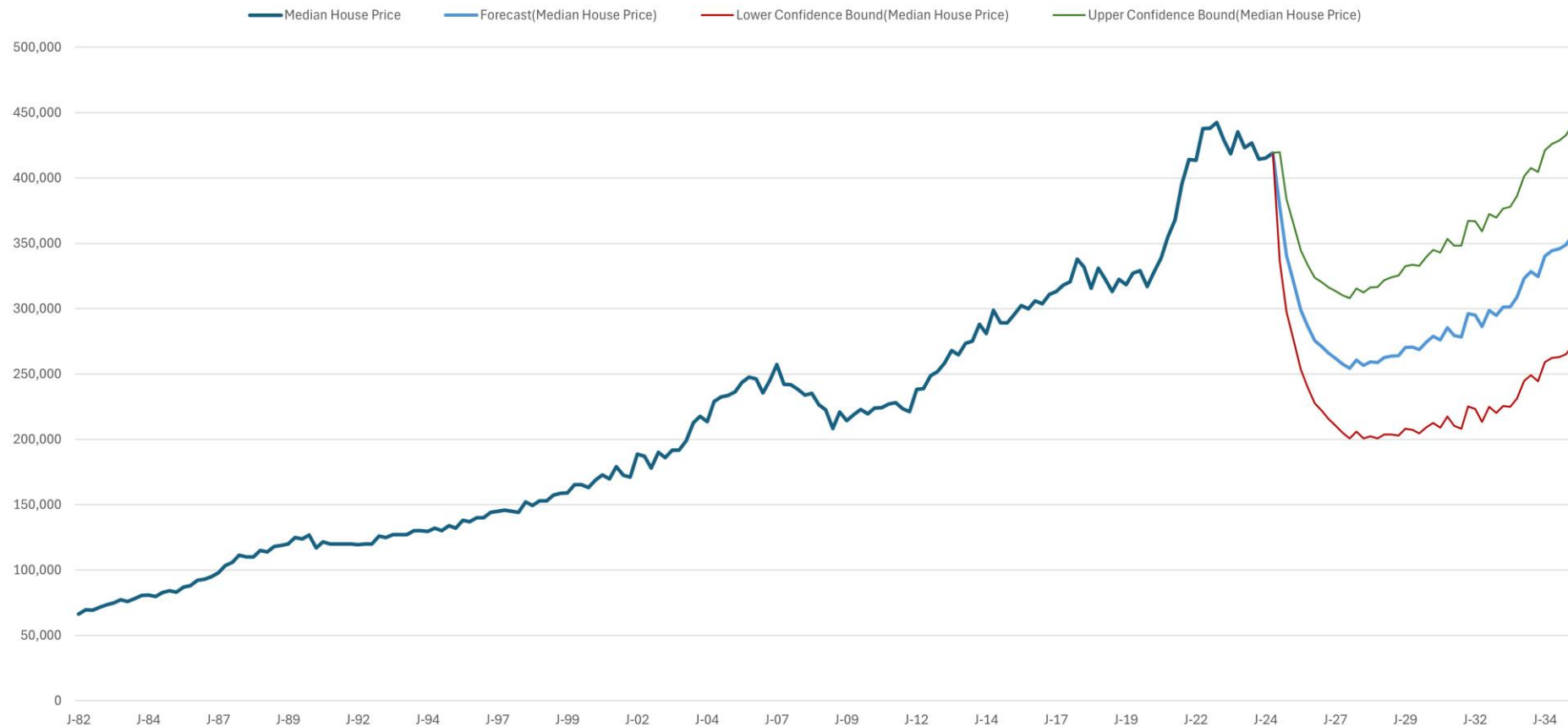
Regression Continued

- ▶ While the original regression using level data yielded a high R^2 of 0.95, this was likely due to **shared trends** across non-stationary variables, leading to a **spurious relationship**.
- ▶ After applying the Dickey-Fuller test, We found that the variables were non-stationary, so we transformed the data using **log-differencing**. This led to a lower R^2 (~ 0.13), which is **normal** in models of changes rather than levels.
- ▶ The lower R^2 reflects the **reduction of artificial trend correlations** and a shift to modeling short-term fluctuations, which are more variable and harder to predict.
- ▶ Therefore, although the R^2 is lower, the model is **statistically more reliable**.

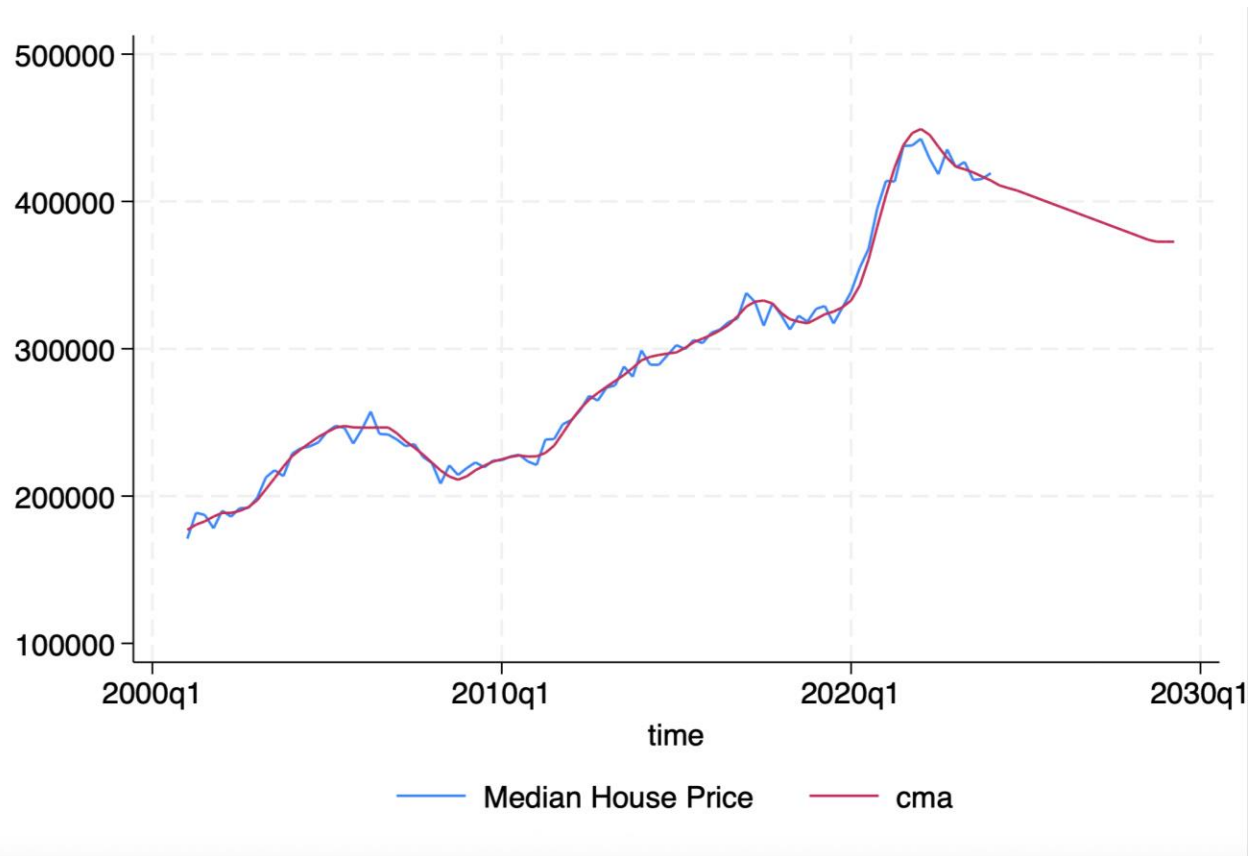


Forecasting

95 Percent Confidence Interval Forecast

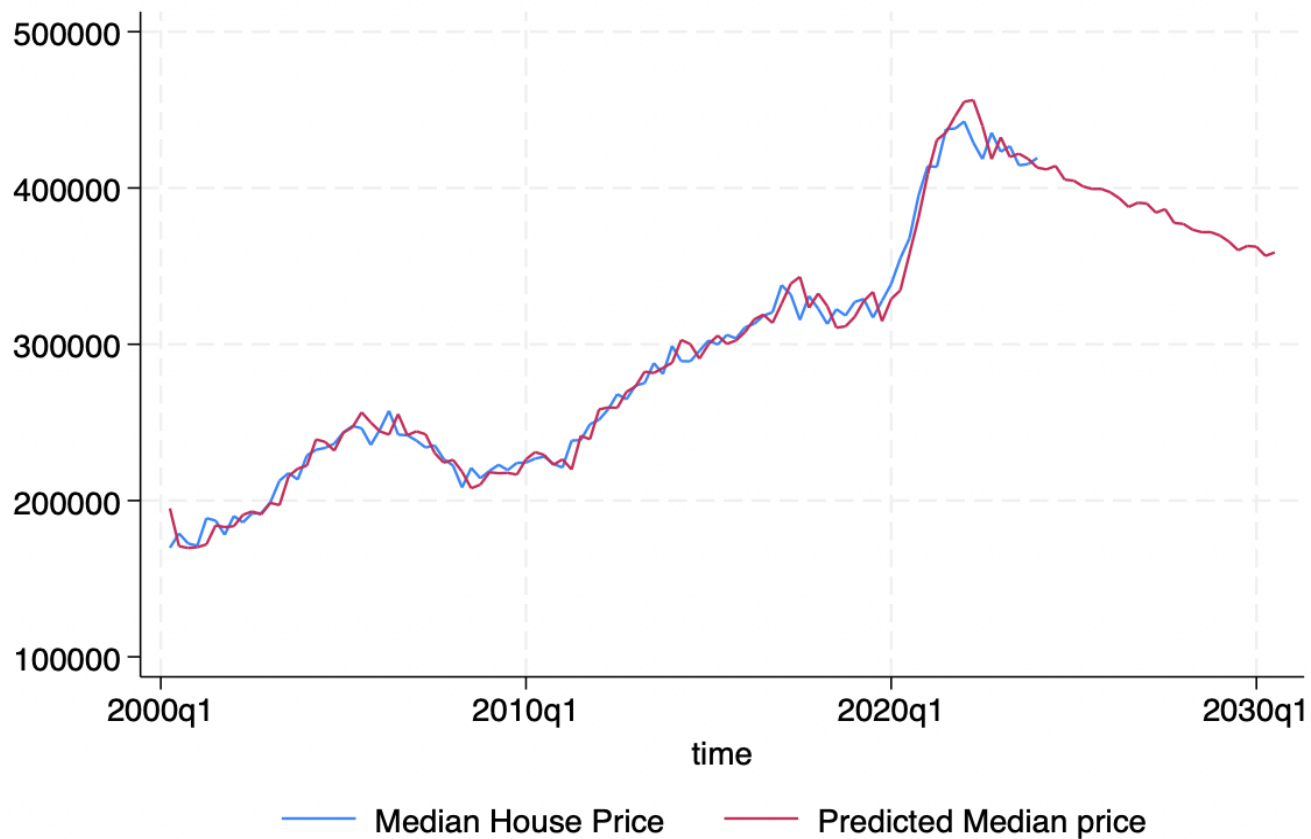


Data range: Jan 1982-2024



Data range: 2000 to 2024

Double Exponential Smoothing with a Centered Moving Average



Data range: 2000 to 2024

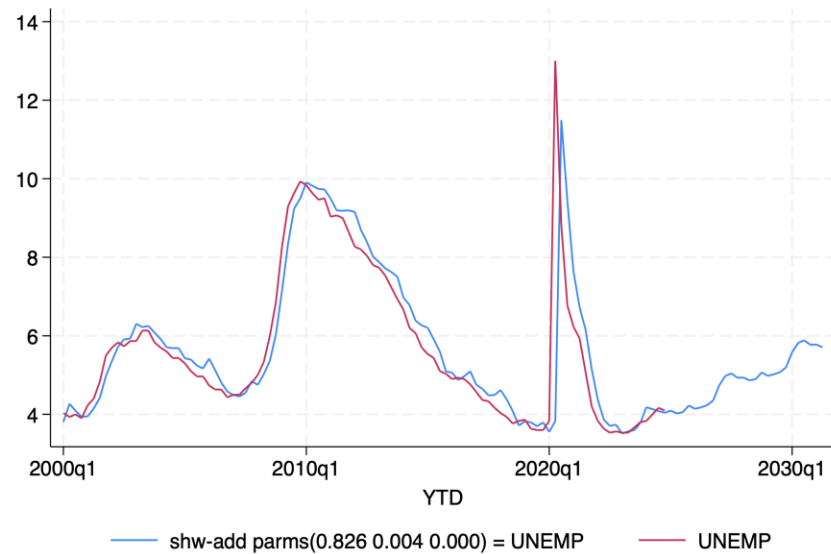
Holt Winters Additive



Data range: 2000 to 2024

Holt Winters Additive(Log Transformed)

Suggested Causes of Housing Price Decreases





Forecasting Approaches

Naive Approach



The naive approach simply uses the previous value as the next forecast. The naive approach has validity but is very simple not showing the trend. Looking at figure



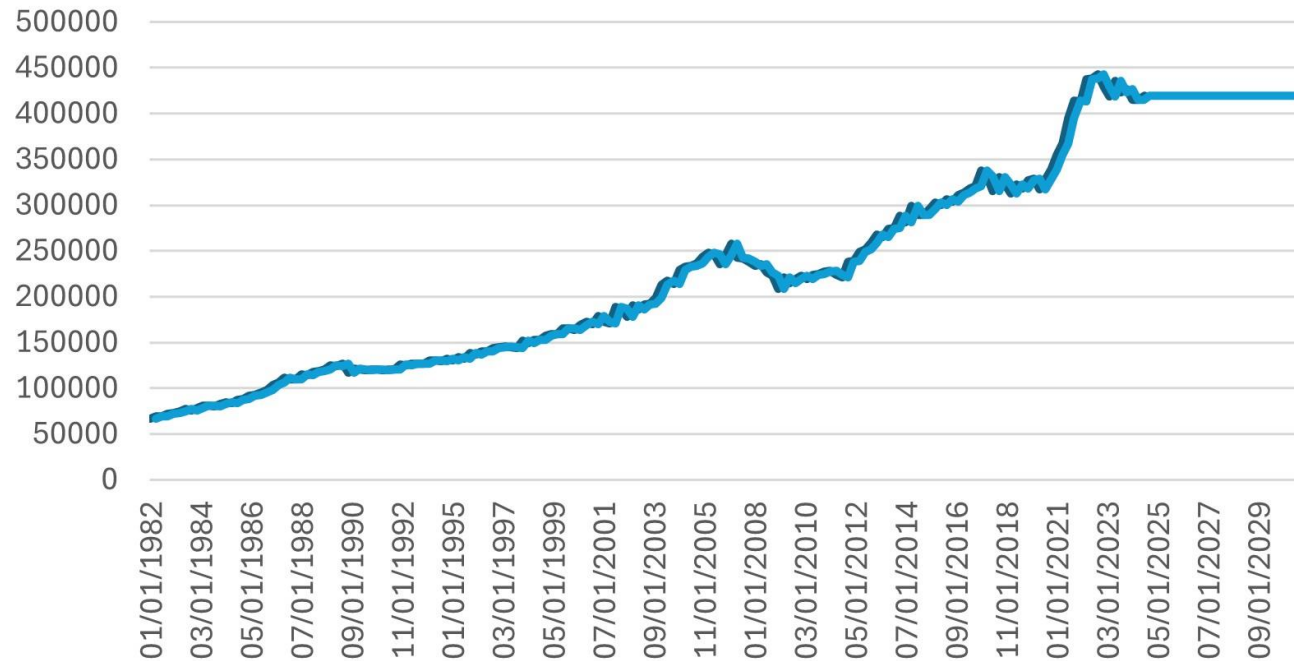
The season naïve captures a seasonal trend to the data but still provided a repetitive result. Although these forecast methods are very simple, they proved grounds for more complex forecasting



The forecast simply cuts off at a certain value and stays level

Naïve and Seasonal Naïve

Naïve



Seasonal Naïve



Forecasting Approaches cont.

► Exponential smoothing

- $\hat{y}_{t+1}|_t = \alpha Y_t + \alpha(1 - \alpha)Y_{t-1} + \alpha(1 - \alpha)^2 Y_{t-2} + \dots$
- A verbal description of this expression is:
- New mean = old mean + (difference between new observation and old mean) / new sample size (= n + 1)

► Moving average

- ◀ $MA^{(2,1,1)}_t = (1/4) \times (Y_{t-2} + Y_{t-1} + Y_t + Y_{t+1})$
- ◀ $MA^{(1,1,2)}_t = (1/4) \times (Y_{t-1} + Y_t + Y_{t+1} + Y_{t+2})$
- ◀ $CMA_t = (1/2) \times [MA^{(2,1,1)}_t + MA^{(1,1,2)}_t]$

Forecasting Approaches cont.

▶ Naive

▶ $\hat{y}_{t+h|t} = y_t$

▶ Drift

▶ $\hat{y}_{t+h|t} = y_t + (h / (T - 1)) \times \sum_{t=2}^T (y_t - y_{t-1})$

▶ $= y_t + h \times ((y_t - y_1) / (T - 1))$

▶ Double exponential smoothing

▶ $S_{2,t} = \alpha \times S_{1,t} + (1 - \alpha) \times S_{2,t-1}$

▶ Holt-winters Additive

▶ $L_t = L_{t-1} + T_{t-1} + \alpha e_t$

▶ $T_t = T_{t-1} + \alpha \beta e_t$

▶ $S_t = S_{t-m} + \gamma e_t$

Stata Code Representation

- ▶ `tsmooth hwinters sm1 = Y, forecast(3)`
- ▶ `tssmooth ma madxpmhp = dexp2Mhp, window(2 1 1)`
- ▶ $\text{madxpmhp}(t) = (1/4) * [x(t-2) + x(t-1) + x(t) + x(t+1)]$
- ▶ `tssmooth ma ma2xpmhp = dexp2Mhp, window(1 1 2)`
- ▶ $\text{ma2xpmhp}(t) = (1/4) * [x(t-1) + x(t) + x(t+1) + x(t+2)]$
- ▶ `tssmooth shwinter ASHW_MHP = Median hp, period(x)`
`additive replace forecast(x)`

Conclusion

- ▶ Price trends provides insight into consumer speculation and consumer confidence
- ▶ Housing prices can serve as a benchmark for the U.S. economy, and it can be a predictive tool
- ▶ Housing market is more than just supply and demand

Citations

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- ▶ Hirano, Tomohiro, and Joseph E. Stiglitz. "Growth and Fluctuations Economies with Land Speculation." *NBER*, 24 Mar. 2025, www.nber.org/papers/w33589.
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